

## DATA SHEET

### PERFORATED DRAIN OR PERFORATED AND FILTERED DRAIN

**PRODUCT DESCRIPTION :** Perforated flexible single wall pipe, with corrugated interior and exterior walls for drainage application use.

**FUNCTION :** Used to collect surface runoff and control the ground water level.

**MANUFACTURING STANDARD :** BNQ 3624-115 Type 2, Type 3\*, compliant to ASTM F667 and meets the OPSS 1840 requirements.

**RAW MATERIAL :** Made from high-density polyethylene (HDPE) complies with properties classification of ASTM D3350 standard.

**TECHNICAL DATA :** Diameter : 50 mm (2 in) to 250 mm (10 in)  
 Length : 3 m (9.8 ft) to 1200 m (3,937 ft)  
 Strength in compression: 210 kPa  
 Manning : 0.015 to 0.017  
 Perforations\*\* : Type 2, 1.8 mm sluice  
                           Type 3, 3.0 mm sluice  
                           Mega 3 (Type 3), more than 3.0 mm sluice  
 Filter : Needle punched nonwoven TXC-10 geotextile, FOS 100 µm  
           Needle punched nonwoven TXC-250 geotextile, FOS 250 µm  
           Woven, FOS 450 µm  
 Maximum backfill depth : 3 m (9.8 ft)



TYPE 2 perforated DRAIN



TYPE 3 perforated DRAIN



TYPE 2 perforated DRAIN  
filtered with TXC-10



TYPE 3 (MEGA 3) perforated DRAIN  
filtered with TXC-250

**AVAILABLE COUPLERS :** Soil tight: see technical data table

### TECHNICAL DATA TABLE

| Diameter |    |             |     |          |      | Length                 | Manning                               | Type  |   | Filter FOS |        |        | HDPE soil tight connectors |    |     |    |
|----------|----|-------------|-----|----------|------|------------------------|---------------------------------------|-------|---|------------|--------|--------|----------------------------|----|-----|----|
| Nominal  |    | Interior    |     | Exterior |      |                        |                                       |       |   |            |        |        |                            |    |     |    |
| mm       | in | mm          | in  | mm       | in   |                        |                                       |       |   |            |        |        |                            |    |     |    |
|          |    |             |     |          |      | m                      | ft                                    | n     | 2 | 3          | 100 µm | 250 µm | 450 µm                     | IC | DBS | SC |
| 50       | 2  | Voir note 2 |     |          |      | 250                    | 820,2                                 | 0,015 | X |            | X      |        |                            |    |     |    |
| 75       | 3  | 73          | 2,9 | 91       | 3,6  | 30,5                   | 100                                   | 0,015 | X |            |        |        | X                          |    | X   |    |
| 100      | 4  | 99          | 3,9 | 116      | 4,6  | 3/15/30/45/75/600/1200 | 9,8/49,2/98,4/147,6/246,1/1968,1/3937 | 0,015 | X | X          | X      | X      | X                          | X  | X   |    |
| 150      | 6  | 150         | 5,9 | 176      | 6,9  | 30/300                 | 98,4/984,3                            | 0,016 | X | X          | X      | X      | X                          | X  | X   | X  |
| 200      | 8  | 198         | 7,8 | 235      | 9,3  | 6/30/120               | 19,7/98,4/393,7                       | 0,016 | X | X          | X      | X      | X                          |    | X   | X  |
| 250      | 10 | 250         | 9,8 | 295      | 11,6 | 6/30/120               | 19,7/98,4/393,7                       | 0,017 | X |            | X      | X      | X                          |    | X   | X  |

Note 1: Values in the table are approximate and may change without notice.

Note 2: Values are available on request.

#### LEGEND

IC: internal coupler

DBS: double bell snap

SC: split coupler

\*The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

\*\* Several factors enter into the selection of the product to be used. Therefore Soleno recommends a granulometric analysis before undertaking a drainage project in order to choose the right pipe according to the type of soil.

## DATA SHEET

### PERFORATED DRAIN OR PERFORATED AND FILTERED DRAIN (CONT'D)

#### FILTER PROPERTIES TABLE

|              | Properties                        | Test method        | FOS 100 µm                  | FOS 250 µm                  | FOS 450 µm |
|--------------|-----------------------------------|--------------------|-----------------------------|-----------------------------|------------|
| Mechanical   | Tensile strength (N)              | CAN-148.1 - No 7.3 | 95                          | 200                         | n/a        |
|              | Elongation at break (%)           | CAN-148.1 - No 7.3 | Min.:65, Max.:105           | Min.:65, Max.:100           | n/a        |
| Hydraulic    | Permeability (cm/s)               | CAN-148.1 - No 4   | 0,15                        | 0,46                        | 0,4        |
|              | Permittivity (S <sup>-1</sup> )   | CAN-148.1 - No 4   | 2                           | 8                           | >2,4       |
|              | Filtration opening size (FOS, µm) | CAN-148.1 - No 10  | 110                         | 250                         | 450        |
|              | Wettability (cm)                  | CAN-4.2 - No 26.3  | <1,0                        | <1,0                        | n/a        |
|              | Flow rate (l/s/m <sup>2</sup> )   | ASTM D4491         | 131                         | 270                         | 238        |
| Construction | Raw material                      |                    | Polyester                   | Polypropylene               | Polyester  |
|              | Manufacturing process             |                    | Non-woven<br>needle punched | Non-woven<br>needle punched | Knit       |

**APPLICATIONS :** Trench drain  
Foundation and basement drainage  
Agricultural subsurface drainage  
Turf and recreation drainage  
Drainage collector and outlet

**OPTION:** Specific perforations are available upon request (minimum volume required).

**ACCESSORIES :** Consult the Drain Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of drain accessories.